

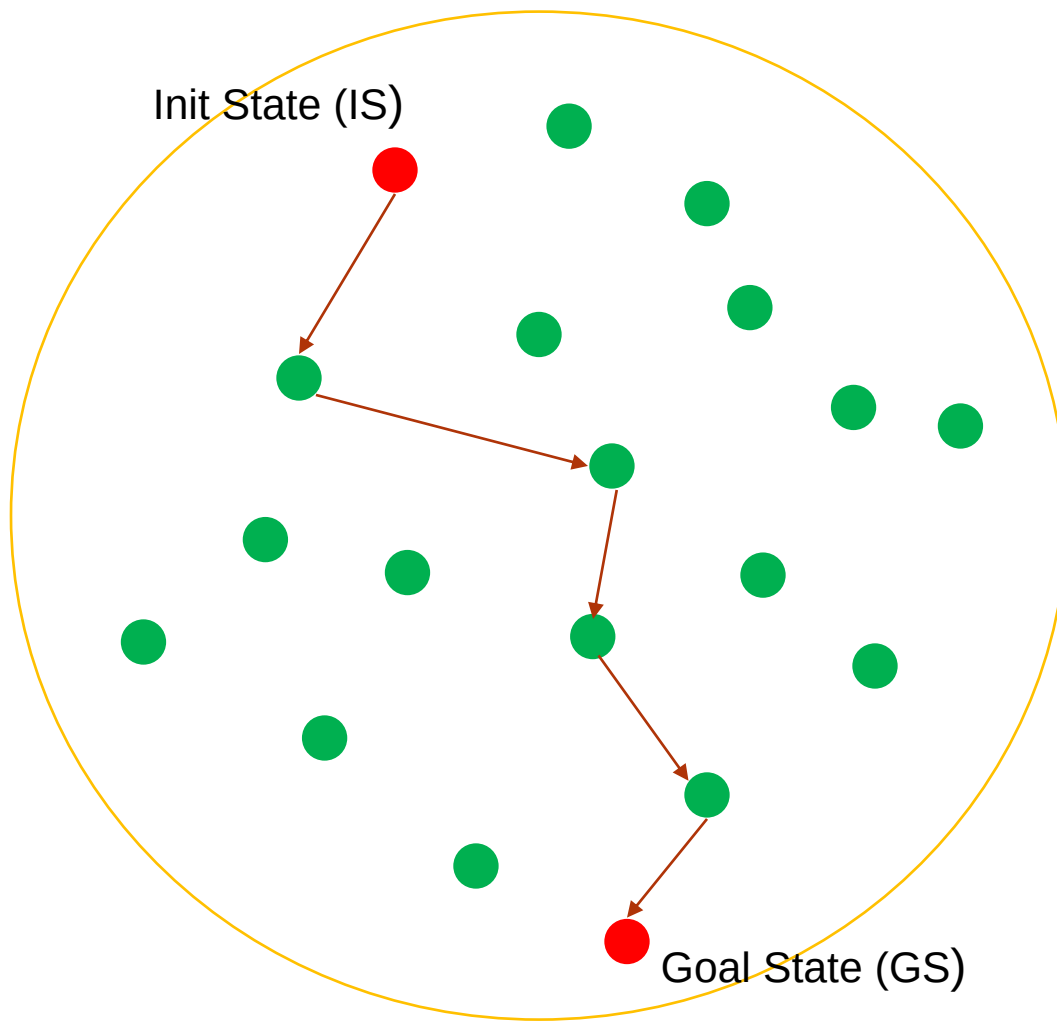
AI problem solving through searching

AI problem solving techniques:

1. **Problem Solving by Searching**
(tick-tac-toe, 8-puzzle, chess)
2. **Problem Solving by Reasoning/ Inferencing**
(Diagnosis, fault-finding)
3. **Problem Solving by Matching**
(face recognition, content similarity)

Defining an AI Problem as **State Space Search**:

1. Define a **State Space** containing all possible configurations (states) of the relevant objects associated with the problem.
2. Specify the **Initial State** within the state space from which problem solving may begin.
3. Specify the **Goal State** which represents an acceptable solution (where the search will terminate)
4. Specify a **set of rules** that describe permissible actions.



State Space vs Search Space

State Space:

All possible states that may arise for the problem under consideration.

Ex: In chess game, the state space would include all possible board configurations/situations (wherever a piece can be placed) during the game.

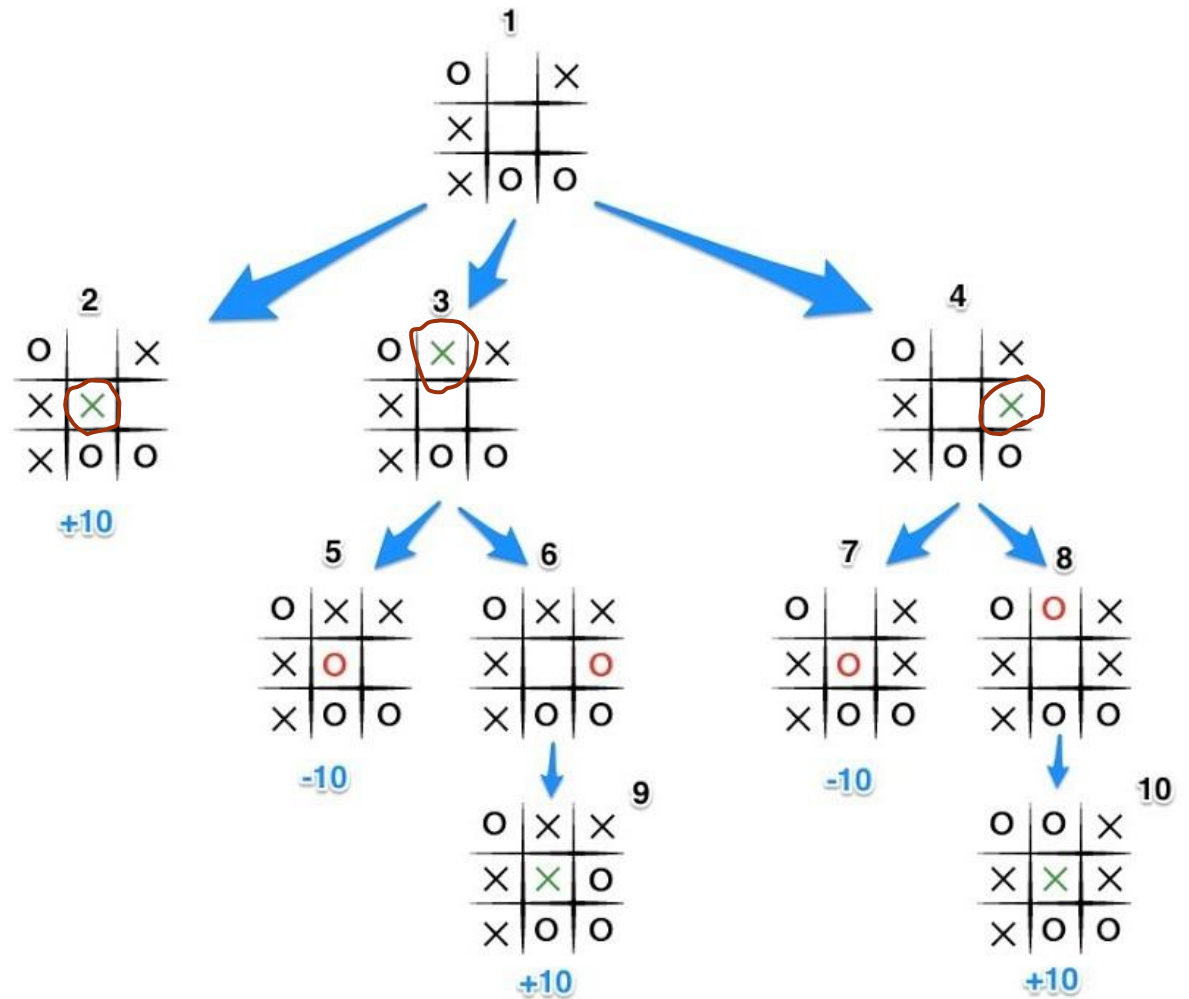
Search Space:

It is a subset of the state space that is actually explored/navigated during the problem-solving process while trying to find a solution.

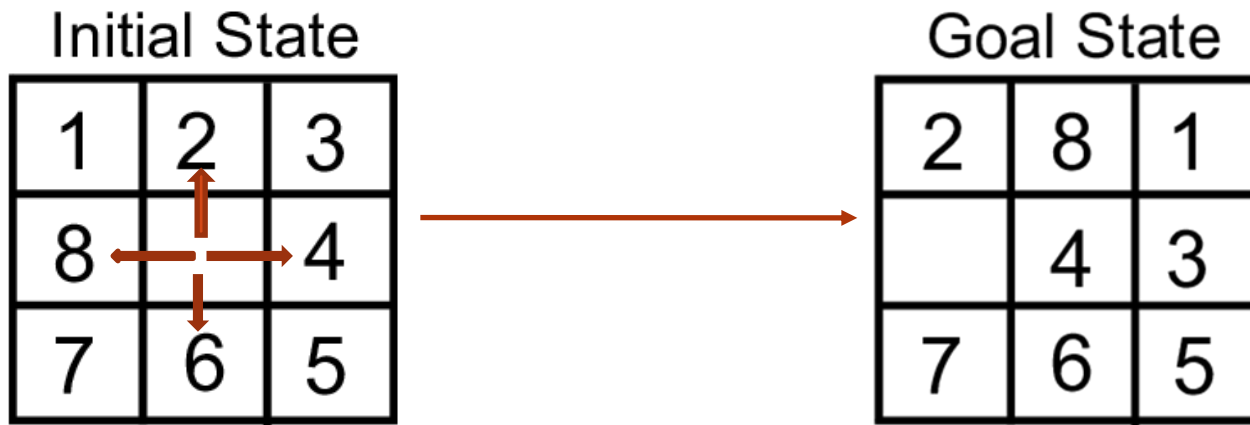
Ex: Only those board configurations/states that are examined during the search for a solution (the best move).

Tic Tac Toe

O	X	
O	X	O
X	O	X



8 Puzzle problem



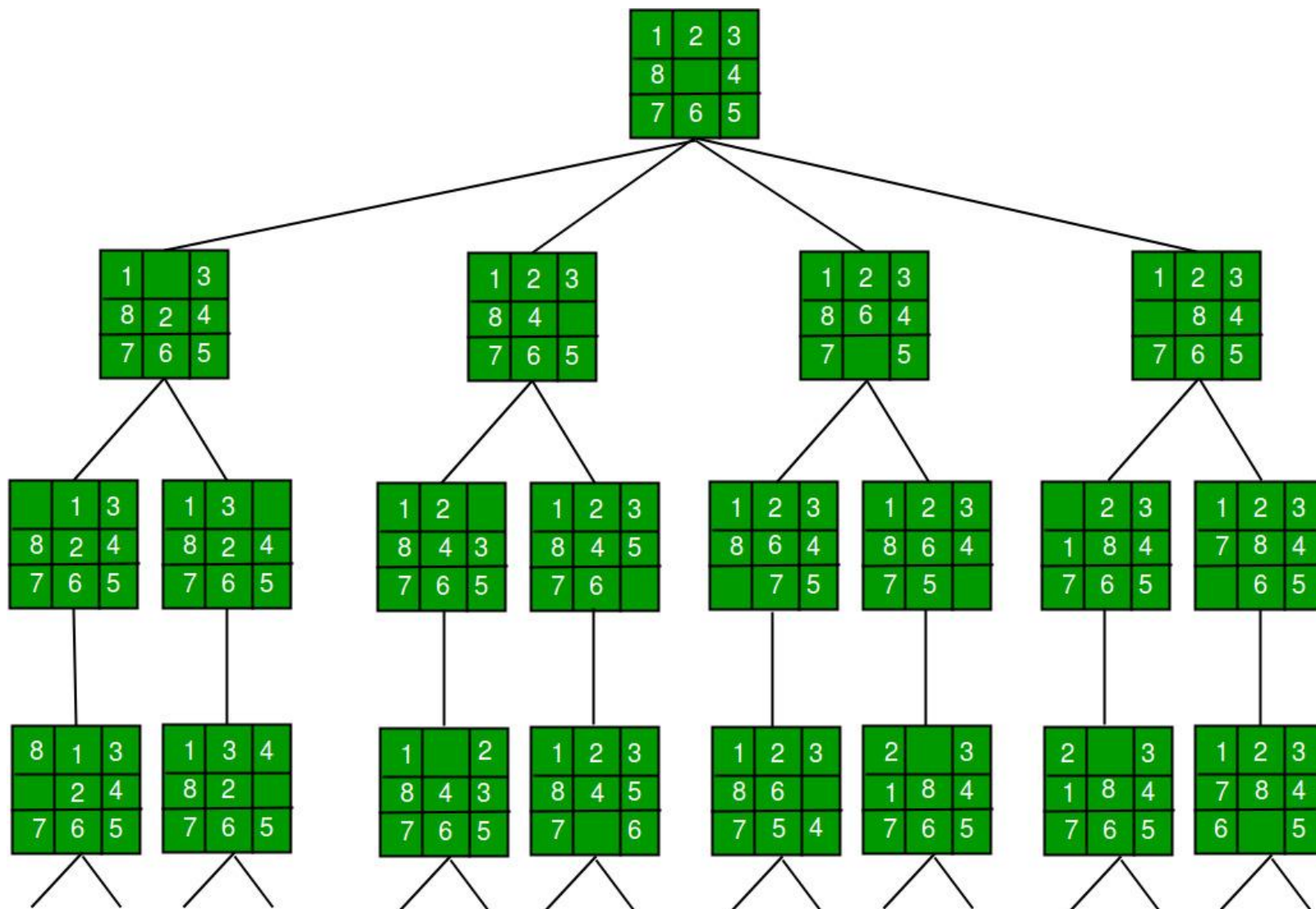
Operators:

Move UP

Move Down

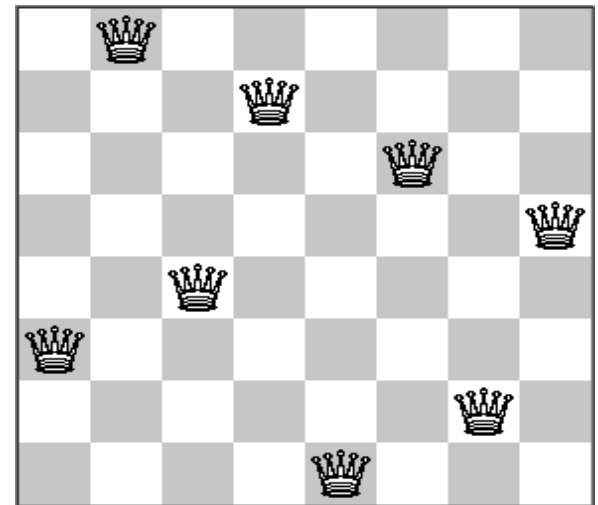
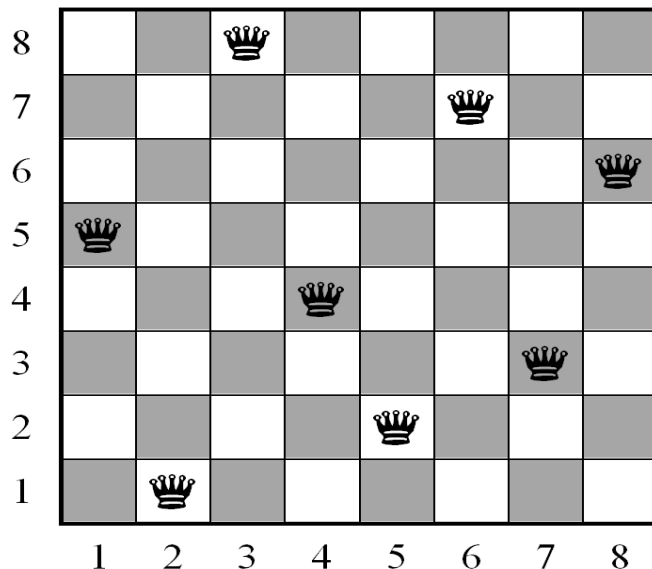
Move Left

Move Right



8 Queen problem

The goal of the 8-queens problem is to place eight queens on a chessboard such that no queen attacks any other queen on the chessboard. (A queen attacks any piece in the same row, column or diagonal.)



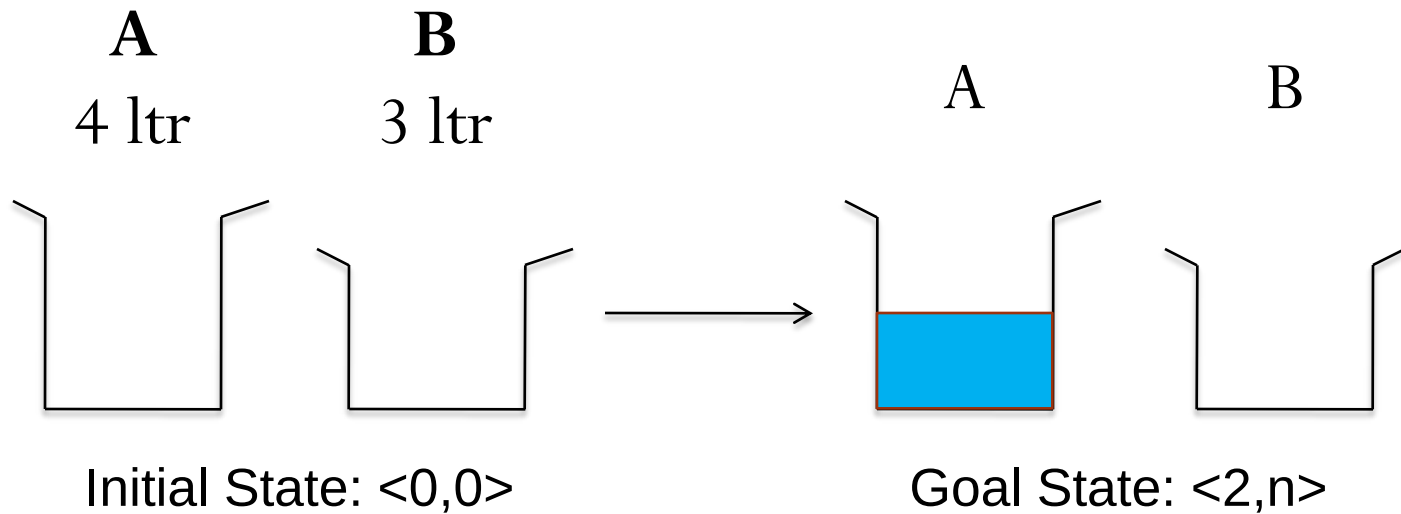
Control Strategy

- Control Strategy in AI refers to a strategy that helps one to select a rule that can be applied at every step while searching for a desired goal state in a state space.
- The objective is to design an efficient control strategy to guide the search process in a state space for finding a solution.

Characteristics of a good control strategy

- A control strategy must cause motion (move from one state to another) otherwise a soln. can never be reached.
- A control strategy must be systematic.
This emphasizes on Local motion & Global motion

A Water Jug Problem



State Space: Described as $\langle x, y \rangle$ ordered pair

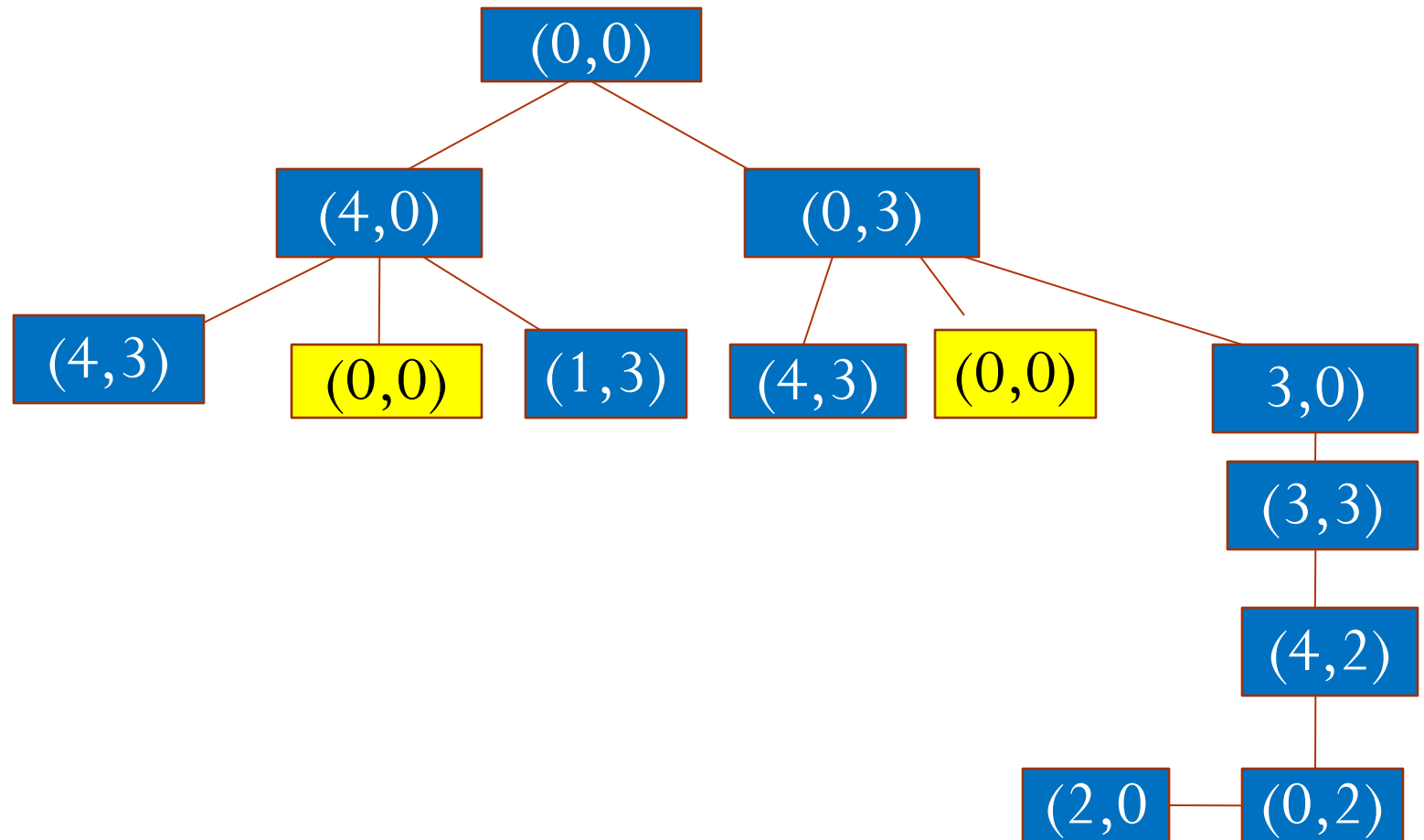
$X = \{0, 1, 2, 3, 4\}$

$Y = \{0, 1, 2, 3\}$

Modeling Water jug problem as a state space search problem

- Construct a tree with the Initial state(IS) as its Root.
- Generate all possible child nodes of the root by applying appropriate rules to the IS.
- For each child node generate further child nodes by applying suitable rules at that stage.
- Continue the process till the goal state is reached.

State Space for the Water jug problem



Creating a rule base:

	<u>L.H.S of rule</u>	<u>R.H.S value</u>	<u>Action</u>
R1:	$\langle x, y \rangle$ if $x < 4$	$\longrightarrow \langle 4, y \rangle$	Fill A
R2:	$\langle x, y \rangle$ if $y < 3$	$\longrightarrow \langle x, 3 \rangle$	Fill B
R3:	$\langle x, y \rangle$ if $x > 0$	$\longrightarrow \langle x - d, y \rangle$	Pour some water out of A
R4:	$\langle x, y \rangle$ if $y > 0$	$\longrightarrow \langle x, y - d \rangle$	Pour some water out of B
R5:	$\langle x, y \rangle$ if $x > 0$	$\longrightarrow \langle 0, y \rangle$	Empty A on ground
R6:	$\langle x, y \rangle$ if $y > 0$	$\longrightarrow \langle x, 0 \rangle$	Empty B on ground

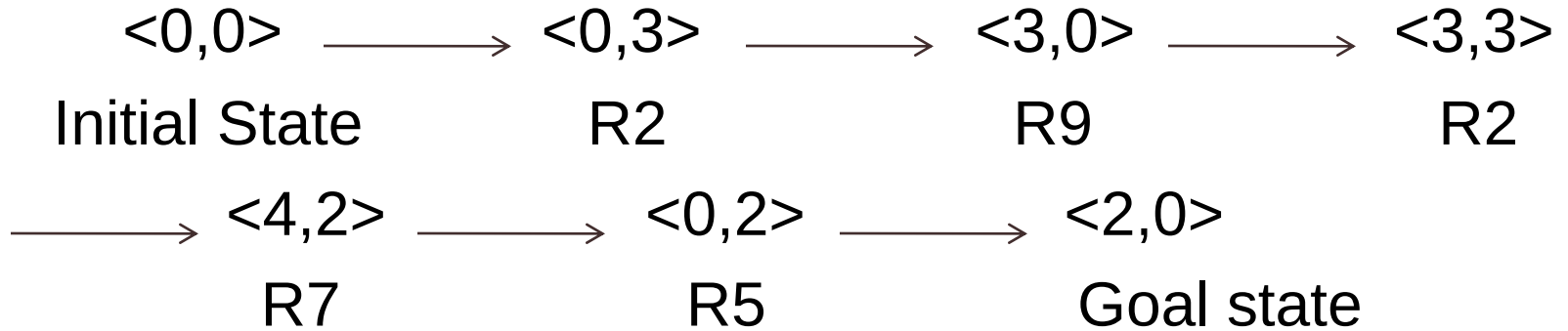
L.H.S of rule

R.H.S value

Action

R7:	$\langle x, y \rangle$ if $x+y \geq 4$ & $y > 0$	$\longrightarrow \langle 4, y-(4-x) \rangle$	Pour from B into A until A is full
R8:	$\langle x, y \rangle$ if $x+y \geq 3$ & $x > 0$	$\longrightarrow \langle x, (3-y), 3 \rangle$	Pour from A into B until B is full
R9:	$\langle x, y \rangle$ if $x+y \leq 4$ & $y > 0$	$\longrightarrow \langle x+y, 0 \rangle$	Pour all from B into A
R10:	$\langle x, y \rangle$ if $x+y \leq 3$ & $x > 0$	$\longrightarrow \langle 0, x+y \rangle$	Pour all from A into B
R11:	$\langle 0, 2 \rangle$	$\longrightarrow \langle 2, 0 \rangle$	Pour 2lts from B to A
R12:	$\langle 2, y \rangle$	$\longrightarrow \langle 0, y \rangle$	Pour 2lts in A on the ground

Possible Rule Sequence:



Problem Solving Strategy:

1. Define problem's state space including initial & goal states.
2. Define a set of valid operators to move in the state space.
3. Search for a path from Initial state to Goal state.

“Search” forms the core of solving AI Problems.

“Effort is to search quickly”.